INSIDE THE EARTH

Reading global earthquakes through space, time, depth and magnitude

DSC5002 Project Team | City University of Hong Kong | USGS 2000–2025, preferred M ≥ 5

46,463
unique earthquake records

7.45×
100-km share / equal-area reference

11.76%
deep events in Tonga / Kermadec / Fiji

28
events with magnitude ≥ 8

01 SPACE & REFERENCE

Question, data and approach

Where are earthquakes concentrated, and what changes when we examine time and depth? We connect five research questions using one frozen, earthquake-only catalogue.

Methods. Unique IDs; UTC 2000–2025; M ≥ 5. Six geographic boxes plus Other. Distances to PB2002 minor great-circle arcs (R = 6,371 km); no automatic removal of extreme or negative depths.

Global geography

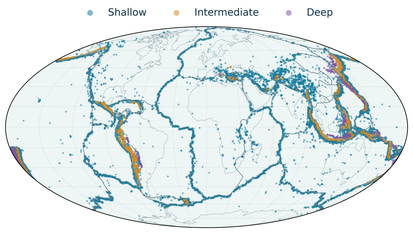


Fig. 1. All 46,463 epicentres on an equal-area Mollweide map. Colours: shallow, intermediate and deep; grey lines: PB2002. Clusters follow the circum-Pacific belts; overlapping points are not an area-normalized density.

Concentrated relative to what?

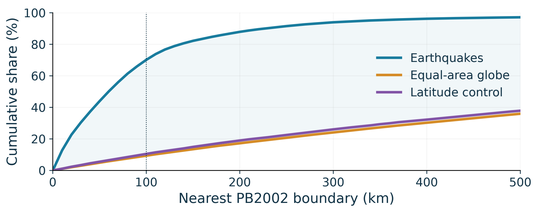


Fig. 2. Within 100 km: **70.15%** of events versus **9.42%** of equal-area globe points; latitude control **10.50%**. Four 16,384-point designs give 9.38–9.53% (numerical sensitivity, not a CI).

Regional proximity is uneven

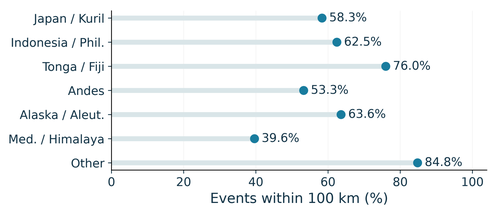


Fig. 3. Each region has its own denominator. Other is a heterogeneous residual category. Reference points cover land and ocean; this is a descriptive geographic baseline, not a seismicity null model.

02 TIME & COMPARABILITY

A fluctuating recorded history

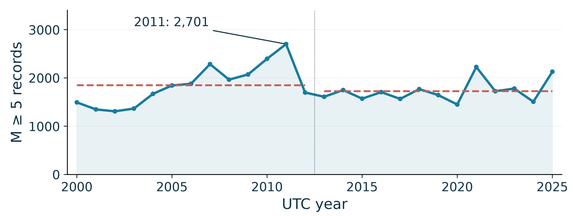


Fig. 4. Annual counts; dashed lines are 13-year means. The 2011 peak is **2,701**. Later-period rates are **6.63% lower at M ≥ 5** and **15.42% lower at M ≥ 6**, after allowing for 4,749 versus 4,748 days.

The direction depends on region

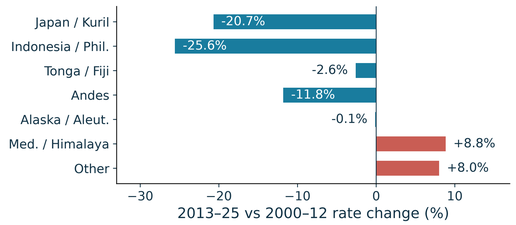


Fig. 5. 2013–2025 versus 2000–2012: Indonesia/Philippines **−25.58%**, Mediterranean/Himalaya **+8.82%**. Region boxes differ in area; these are within-region changes, not comparisons of hazard.

Can the periods be compared?

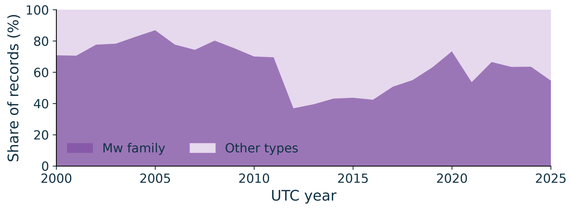


Fig. 6. Mw-family share shifts from **73.45% to 55.05%** between periods. Preferred magnitude scales are mixed; selecting Mw records alone does not homogenize the catalogue.

Reporting and sensitivity
Horizontal-error availability: **0.28%** → **86.80%**. Missing does not mean zero error.
Removing 30-day / 500-km windows around 28 M ≥ 8 events changes the M5 contrast to **−3.37%**. This is a window sensitivity, not declustering.

03 DEPTH & RELATIONSHIPS

Similar medians, different tails

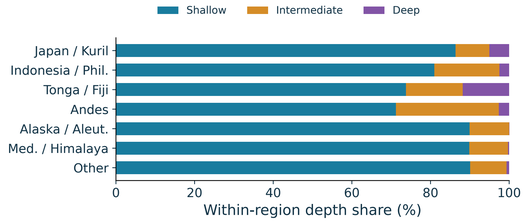


Fig. 7. Globally **83.20%** are shallow (<70 km). Japan/Kuril and Tonga/Fiji both have median depth 33 km, but deep-event shares are **5.01% and 11.76%**. Andes has **26.19%** intermediate events.

Local sections reveal structure

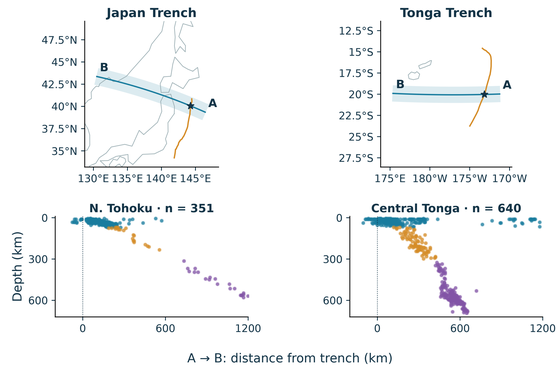


Fig. 8. Maps: gold trench, blue route/swath, star at 0 km. A → B: −200 to +1,200 km; ±100 km swaths, equal kilometre scales. Deep events: **21/351** in northern Tohoku; **186/640** in central Tonga. Routes were selected after exploration.

Pooled and regional associations

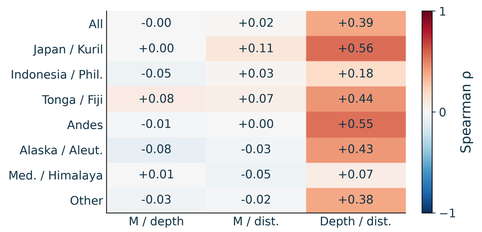


Fig. 9. Pairwise Spearman ρ, all eligible records. Depth–distance association is +0.39 overall and +0.56 in Japan/Kuril; magnitude associations are weak. Correlation does not identify causality.

04 MAGNITUDE & SEQUENCES

Large events are much rarer

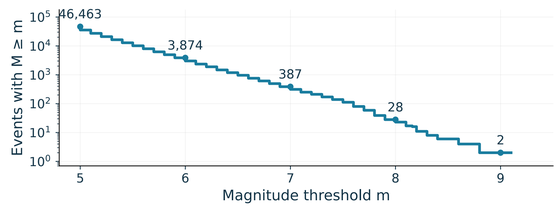


Fig. 10. Inclusive N(M ≥ m), log count axis: **3,874** records ≥6; **387** ≥7; **28** ≥8. The M5 extraction threshold does not establish catalogue completeness; no predictive b-value is fitted.

Counts are not an energy scale

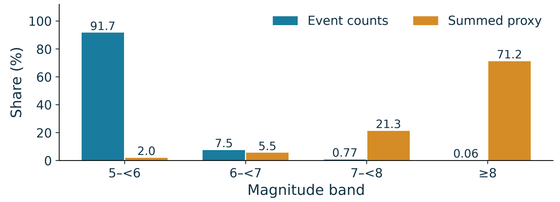
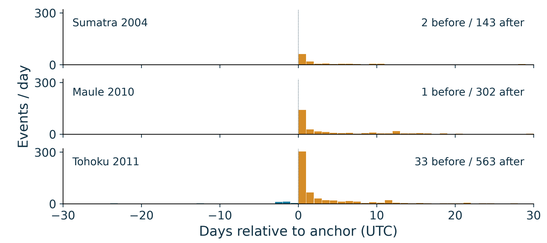
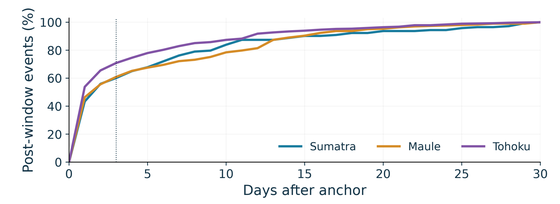


Fig. 11. M ≥ 8 accounts for **0.06%** of events but **71.21%** of the summed illustrative proxy, $R(M)=10^{1.5(M-5)}$. A unit increase multiplies R by 31.6. Mixed scales: not measured joules, damage or risk.

Sequences through a fixed window



Figs. 12–13. Within 500 km and ±30 days; anchors excluded. Gold: post-event; blue: pre-event. First three days contain **60.1%, 60.9%, 70.9%** of post-window counts. Each cumulative curve uses its own 30-day total as denominator.



CONCLUSION Strong boundary concentration, contrasting depth tails and clustered sequences emerge. Their interpretation depends on the geographic reference, catalogue composition and window definition; none estimates future hazard.

[1] USGS FDSN Event Service · [2] Bird (2003), PB2002; GeoJSON (ODC-By) · [3] Natural Earth 110m (public domain) · [4] USGS magnitude–energy guidance. Full methods and sensitivity results: project repository.